Chapter 1

The Special Theory of Relativity

1.1 Introduction

The development of physical knowledge is divided, somewhat arbitrarily, into two periods. The pre-20th century physics is generally called *classical* while the 20th century physics is known as *modern physics*. The difference, however, is not so simple.

Modern physics rests on two new concepts: *relativity* and *quantum mechanics*. While some of the classical results may appear as special cases of relativistic laws, the results of quantum mechanics have no classical analogue. Together these two theories have completely revolutionized our view of the structure of matter and the nature of radiation and the interaction between them.

1.2 The Ether Hypothesis

Newton's mechanics, which embodies the concept of inertia introduced by Galileo, incorporates the law that a moving body subject to no forces will continue to move with a constant speed in a fixed direction($\mathbf{F} = m\frac{d\mathbf{v}}{dt} = 0 \Rightarrow \mathbf{v} = const$).

The motion of a body, to be meaningful, has always to be referred to some type of a coordinate system which acts as a frame of reference and may itself be in a motion. A frame of reference (or coordinate system) in which Newton's laws are valid is called an *inertial frame* or *Galilean-Newtonian frame*. Such asystem has no acceleration relative to a frame of reference which is *absolutely fixed*. This raises the question: does an absolutely fixed frame of reference really exist? The earth,the sun,the starts etc are excluded because they are in constant motion. In the Infinite regress noting appears to be absolutely at rest which could serve as a fixed frame of reference.This makes it impossible to check the validity of Newton's law. It points to a serious flaw in the logical structure of classical physics.

Again, it is known that light from the sun and the stars reaches the earth after traveling millions of miles in empty space. Maxwell found it difficult to imagine howan electromagnetic field could propagate as a self-supporting entity in vacuum.It was argued that just as sound waves propagate by setting into vibration the medium through which they travel and water in essential for the propagation of water waves,analogously light waves require a medium or a carrier in which the electric and magnetic fields could reside

and in which the waves could push themselves forward. This hypothetical medium was given the name *luminiferous ether*.

It seemed that the ether hypothesis not only provided on idea medium through which Maxwell's electromagnetic waves could propagate, it also provided the primary coordinate system which was absolutely at rest and relative to which Newtonian mechanics could hold true.

According to the classical law of addition of velocities, an observer on earth should find a light signal to travel faster with the ether wind than against it. By the same token a light signal at right angles to the ether-wind should be faster than when traveling against it. In1887, two Americans, A.A. Michelson and E.W. Morley, set out to measure the relative velocity of the earth with respect to the ether with the instrument they had invented (called interferometer.) Their experiment has witnessed umpteen repetitions and numerous refinements.Surprisingly, despite best efforts, the presence of ether could not be detected.

1.3 The Michelson-Morley Experiment

In this experiment, two sets of light waves derived from a common monochromatic source S are made to interfere at a point O after traveling equal distances along (AB) and across (AC). The slit image appears as a bright line if the path difference is a whole number of the wavelengths of light and it is dark line if the two paths differ by an odd number of half wavelengths. These bright and dark fringes cross the field of view if the apparatus is rotated through 90°.

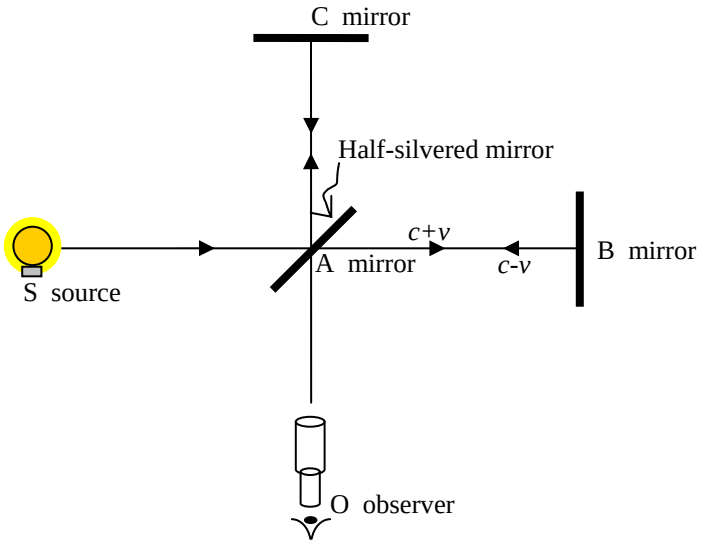


fig. 1.1

Assume the earth is rotated parallel to SB with velocity *v* relative to fixed ether (*earth*→ the river, *ether*→ the bank.)

- Total time (t_1) taken by the ray to cover AB and back BA:

$$t_1 = \frac{L}{c+v} + \frac{L}{c-v} = \frac{2Lc}{c^2 - v^2} = \frac{2L}{c}\left(\frac{1}{1 - v^2/c^2}\right)$$

where $c = 3\times10^8\, m/s$ (speed of light)

- Total time (t_2) taken by the ray to cover AC and back CA:

$$t_2 = \frac{L}{\sqrt{c^2 - v^2}} + \frac{L}{\sqrt{c^2 - v^2}} = \frac{2L}{\sqrt{c^2 - v^2}} = \frac{2L}{c}\left(\frac{1}{\sqrt{1 - v^2/c^2}}\right)$$

Note: $t_1 > t_2$.

$$\therefore \quad \frac{t_1}{t_2} = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{1.1}$$

If we know t_1 / t_2, we can fine *v* from Eq. 1.1.

- The time difference between the two journeys Δt is

$$\Delta t = t_1 - t_2 = \frac{2L}{c}\left(\frac{1}{1 - v^2/c^2} - \frac{1}{\sqrt{1 - v^2/c^2}}\right) \tag{1.2}$$

If we rotate the apparatus by 90°, the new time difference $\Delta t'$ is:

$$\Delta t' = t_1' - t_2' = \frac{2L}{c}\left(\frac{1}{\sqrt{1 - v^2/c^2}} - \frac{1}{1 - v^2/c^2}\right) \tag{1.3}$$

Michelson looked for a shift in the interference pattern when this apparatus was rotated by 90°. The time difference is:

$$\Delta t - \Delta t' = \frac{4L}{c}\left(\frac{1}{1 - v^2/c^2} - \frac{1}{\sqrt{1 - v^2/c^2}}\right)$$

since $v << c$, according to the binomial theorem,

$$(1 \pm x)^n = 1 \pm nx + \frac{n(n-1)}{2!}x^2 \pm \ldots \approx 1 \pm nx, \quad x^2 << 1$$

$$\Delta t - \Delta t' = \frac{4L}{c}\left[\left(1 + v^2/c^2\right) - \left(1 + \tfrac{1}{2}v^2/c^2\right)\right] = \left(\frac{2L}{c}\right)\left(\frac{v^2}{c^2}\right)$$

This will give a path difference $c(\Delta t - \Delta t')$ or a *fringe shiftn* given by:

$$n = \frac{c(\Delta t - \Delta t')}{\lambda} = \frac{2L}{\lambda}\left(\frac{v}{c}\right)^2 \tag{1.4}$$

Michelson and Morely took v = 3 x 10^4 m/s (earth's orbital speed around the sun), λ = 5.9 x 10^{-7} m, L = 11 m. The expected n was 0.4. The interferometer was capable of detecting

shift of $n = 0.01$. To every body's surprise, *no fringe shift whatever was found* ($n = 0$). The failure, therefore, could not be ascribed to experimental errors or lack of sensitivity of the apparatus.

The negative results of M-M experiment had two consequences:

1. The ether has no measurable properties- an end to the ether hypothesis.
2. The speed of light, c, in free space is the same everywhere and is independent of the relative motion of its source and observer.

1.4 The Special Theory of Relativity

The theory of relativity results from an analysis of the physical consequences implied by the absence of a universal frame of reference. The special theory of relativity, developed by Albert Einstein in 1905, treats problems involving the motion of frames of reference at *constant velocity* (i.e., both constant speed and constant direction) with respect to one another; the general theory of relativity, proposed by Einstein a decade later, treats problems involving frames of reference *accelerated* with respect to one another.

The basic postulates of the special theory of relativity are as follows:

1. **The Principle of Relativity**: The laws of physics are the same in all frames of reference moving with constant velocity with respect to one another.
2. **The Constancy of the Speed of Light**: The speed of light in free space, c, is independent of the motion of its source or of the observer and is the same for all observers.

The first postulate expresses the absence of a universal frame of reference. If the laws of physics had different forms for different observers in relative motion, it could be determined from these differences which objects are "stationary" in space and which are "moving." But because there is no universal frame of reference, this distinction does not exist in nature.

The second postulate follows directly from the results of M-M experiment and others.

1.5 The Galilean Transformation

Consider two orthogonal coordinate systems S and S' such that their x-axes are in the same line and S' has a constant linear relative velocity v with respect to S along the positive x-axis. Suppose we make measurements of time from the instant when the origins of S and S' just coincide.

If an event takes place at a position (x, y, z) and time $t$with respect to S (the distance between the origins of S and S' at this instant is vt), the same event will have different set of coordinates (x', y', z') with respect to S'.

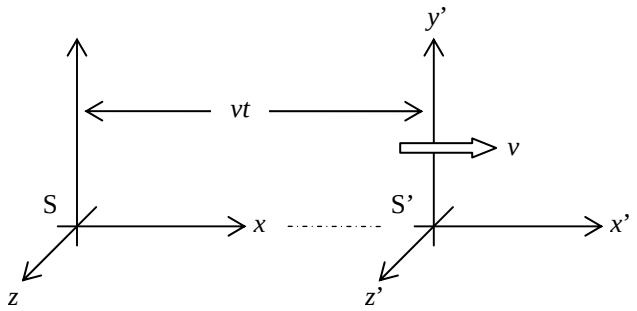


fig 1.2

The relationship between (*x*, *y*, *z*) and (*x*', *y*', *z*') in Galilean transformation is:

$$x' = x - vt \qquad 1.5$$

There is no relative motion in the *y* and *z* directions,

$$y' = y \qquad 1.6$$

$$z' = z \qquad 1.7$$

And from *common sense experience*,

$$t' = t \qquad 1.8$$

The set of equations 1.5-1.8 are known as the *Galilean transformations.*

Differentiating with respect to time,

$$\frac{dx'}{dt'} = v'_x = v_x - v \qquad 1.9a$$

$$v'_y = v_y \qquad 1.9b$$

$$v'_z = v_z \qquad 1.9c$$

Again differentiating with respect to time,

$$\frac{dv'_x}{dt'} = a'_x = a_x \qquad 1.10a$$

$$a'_y = a_y \qquad 1.10b$$

$$a'_z = a_z \qquad 1.10c$$

Thus Newton's law of motion holds in S as well as S':

$$F = ma = ma'$$

The above transformations violate both of the postulates of special theory of relativity.

The first postulate calls for identical equations of physics in S and S', but the fundamental equations of electricity and magnetism assume very different forms when the above transformations are used. The second postulate calls for the same value of the speed of light *c*whetherdetermined in S or S'. But if it is *c* in S, it would be *c*' = *c* – *v* in S'. So adifferent transformation is required which satisfy the postulated of the special theory of relativity.

1.6 The Lorentz Transformation

Assume a more general formulation as:

$$x' = k(x - vt) \tag{1.11}$$

where*k* is a factor of proportionality that does not depend upon either *x* or *t* but may be a function of *v*.

By the principle of equivalence of S and S' (1st postulate):

$$x = k(x' - vt') \tag{1.12}$$

Obviously

$$y' = y, \quad z' = z$$

Hence,

$$x = k[k(x - vt) + vt']$$

$$\Rightarrow t' = kt + \left(\frac{1 - k^2}{kv}\right)x \tag{1.13}$$

The speed of light is the same in S and S':

$$x = ct \tag{1.14}$$

$$x' = ct' \tag{1.15}$$

Substituting the values of *x*' and *t*' in $x' = ct'$:

$$k(x - vt) = c\left[kt + \left(\frac{1 - k^2}{kv}\right)x\right]$$

Substituting $x = ct$:

$$k(c - v)t = c\left[kt + \left(\frac{1 - k^2}{kv}\right)x\right]t$$

$$\Rightarrow k = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{1.16}$$

Therefore the Lorentz transformations become:

$$x' = k(x - vt) = \frac{x - vt}{\sqrt{1 - v^2/c^2}} \tag{1.17}$$

$$y' = y \tag{1.18}$$

$$z' = z \tag{1.19}$$

$$t' = k\left(t - \frac{v}{c^2}x\right) = \frac{t - \frac{v}{c^2}x}{\sqrt{1 - v^2/c^2}} \tag{1.20}$$

The inverse Lorentz transformations are obtained by interchanging the coordinates and replacing *v* by –*v* in the above equations:

$$x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}} \qquad 1.21$$

$$t = \frac{t' + \frac{v}{c^2}x'}{\sqrt{1 - v^2/c^2}} \qquad 1.22$$

From Eq. 1.16, it is evident that for $v << c$, $k \approx 1$ and the Lorentz transformations reduce to the form of Galilean transformation. This indicates that classical mechanics can be regarded as a special case of relativistic mechanics.

1.7 Contraction of Length

Consider two coordinate systems S and S' with their x-axis coinciding at $t = t' = 0$ and S' moving with a uniform relative velocity $v$with respect to S in the positive x-direction. Further, imagine a rod AB at rest relative to S'. The length as measured at rest is called the *proper length.*

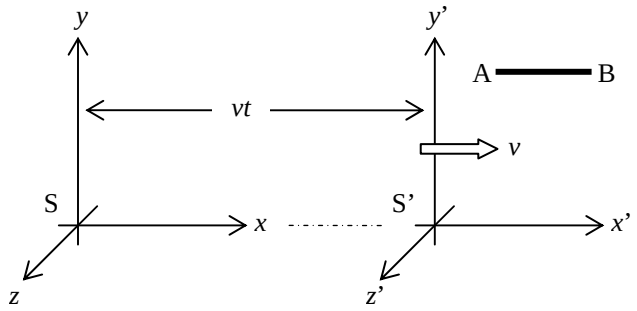


*fig*1.3

In S', $L_o = x_2' - x_1'$ and the rod length when measured relative to S is $L = x_2 - x_1$.

$$\begin{aligned} L_o &= x_2' - x_1' \\ &= k(x_2 - vt) - k(x_1 - vt) \\ &= k(x_2 - x_1) = kL \end{aligned}$$

$$\Rightarrow L = L_o\sqrt{1 - v^2/c^2} \qquad 1.23$$

To an observer in S, therefore, it would appear that the length of the rod (in S') has contracted by the factor $\sqrt{1 - v^2/c^2}$. Again, if the rod is at rest in S, it would appear to an observer in S' as contracted by the same factor. There is no length contraction perpendicular to the relative motion since $y' = y, \quad z' = z$.

These considerations show that space is not a rigid fixed system having an absolute physical significance. It changes in size depending upon the motion of the observer.

Example 1

A rod A'B' inclined at θ_o from the horizontal is at rest in S'. Find the inclination angle as measured by an observer in S.

Example 2

A rocket ship is 100 meters long on earth. In flight its length is 99 meters. Find its velocity.

1.8 Time Dilation

To understand the effect of motion of observers on time intervals, imagine a gun placed at the position (x', y', z') in S' firing two shots at times t_1' and t_2' measuredwith respect to S'. Since the gun is fixed in S', it has velocity $v$with respect to S in the direction of the positive x-axis. Let $t = t_2 - t_1$ represent the time interval between the two shots as measured by an observer in S. In S', in which the gun is at rest, let $t_o = t_2' - t_1'$ be the time interval between the same two shots (called *proper time*) recorded by an identical clock. From Eq. 1.22,

$$t_1 = \frac{t_1' + \frac{v}{c^2}x'}{\sqrt{1 - v^2/c^2}}, \quad t_2 = \frac{t_2' + \frac{v}{c^2}x'}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow t = t_2 - t_1 = \frac{t_2' - t'}{\sqrt{1 - v^2/c^2}}$$

$$\therefore t = \frac{t_o}{\sqrt{1 - v^2/c^2}} \qquad 1.24$$

This implies that to an observer in S it would appear that the interval of time between the firing of the two shots was longer than that measured by an observer in the system S' (moving with the gun). In a similar manner, it can be shown that to an observer in S' this interval would appear to be longer than that measured by an observer in S.

This time dilation effect, then, states that if two observers are moving with constant velocity with respect to each other, it appears to each that the other's time interval have dilated or the other's time process have slowed down. In short *moving clocks run slow*. Note that the proper time is always the smallest possible time interval between two events.

It is interesting here to notice that when time appears slower in a moving system, not only do clocks on that system appear slower but also all time processes such as respiration, digestion, and aging would appear to have slowed down if the biological system is moving with a sufficiently large uniform linear velocity with respect to a measuring clock.

Example 3

Two twins A and B are 20 years old each. B set out on a very high speed spaceship at 0.9$c$to study a nearby star system. While A was celebrating his 50th birthday, B returned back home. How old is B?

Example 4

A μ- meson decays into an electron 2 μs after it is created. Its typical speed is 2.994 x 10^8 m/s. So it travels (2 x 10^{-6}s)(2.994 x 10^8 m/s) = 600 meters before decaying. In nature, μ- mesons are created high in the atmosphere by fast cosmic-ray particles and reach sea level, a much longer distance. How are they able to reach the earth's surface?

1.9 Relativity and Simultaneity

Common sense tells us that if an observer O observes two events to occur simultaneously, another observer O' with a relative motion of uniform translation should also find these events to be simultaneous.

Actually, if O observes two events at (x_1, y_1, z_1) and (x_2, y_2, z_2) simultaneously at time t, then to O' moving with velocity v relative to O, one event appears prior to the other. The classical concept of simultaneity, therefore, requires a modification.

Assume O' observes these events at (x_1', y_1', z_1') and (x_2', y_2', z_2'). Then,

$$x_1' = k(x_1 - vt), \quad x_2' = k(x_2 - vt)$$

so that

$$x_1' - x_2' = k(x_1 - x_2)$$

also

$$t_1' = k\left(t - \frac{v}{c^2}x_1\right), \quad t_2' = k\left(t - \frac{v}{c^2}x_2\right)$$

so that

$$t_1' - t_2' = k(x_2 - x_1)\frac{v}{c^2}$$

$$\therefore t_1' - t_2' = \frac{v}{c^2}(x_2' - x_1') \neq 0 \qquad 1.25$$

This last equation would show that for $v > 0$, the events nearer to O' in space will appear to have happened latter in time. If O' moves in the opposite direction relative to O, the events will appear to have happened in the reverse order. The concept of simultaneity, therefore, is not absolute but entirely relative.

It is evident that the distance between the above events is:

$$\Delta x' = \frac{c^2}{v}\Delta t' \quad \Rightarrow \quad \Delta x' > c\Delta t'$$

This distance is such that no signal can pass between the two events, which is in agreement with the theory of cause and effect, a concept of Newtonian physics which is valid in relativistic physics also.

1.10 Relativistic Law of Addition of Velocities

Consider an object moving with a constant velocity u relative to a frame of reference S along its positive x-axis. Let its velocity relative to another reference system S' be u' where S' has a constant relative velocity v in the same direction with respect to S.

According to the mechanics of Newton$u' = u - v$. But which is incorrect as shown by the M-M experiment. The correct result is obtained using the Lorentz transformations.

Suppose an object moves from $P_1(x_1, y_1, z_1, t_1)$ to$P_2(x_2, y_2, z_2, t_2)$ relative to S. Let P_1 and P_2 relative to S' be (x_1', y_1', z_1', t_1) and (x_2', y_2', z_2', t_2);

$$u = \frac{x_2 - x_1}{t_2 - t_1}, \quad u' = \frac{x_2' - x_1'}{t_2' - t_1'}$$

$$u' = \frac{k(x_2 - vt_2) - k(x_1 - vt_1)}{k\left(t_2 - \frac{v}{c^2}x_2\right) - k\left(t_1 - \frac{v}{c^2}x_1\right)}$$

$$= \frac{(x_2 - x_1) - v(t_2 - t_1)}{(t_2 - t_1) - \frac{v}{c^2}(x_2 - x_1)}$$

$$= \frac{\frac{(x_2 - x_1)}{(t_2 - t_1)} - v}{1 - \frac{v}{c^2}\frac{(x_2 - x_1)}{(t_2 - t_1)}}$$

$$\therefore u' = \frac{u - v}{1 - \frac{uv}{c^2}} \qquad 1.26$$

Similarly, u in terms of u' and v is

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}} \qquad 1.27$$

If $v << c$, the classical equations result.

A More General Way of Deriving the Law of Addition of Velocities:

In differential form, the Lorentz equations are:

$$dx' = k(dx - vdt) \qquad 1.28$$

$$dy' = dy \qquad 1.29$$

$$dz' = dz \qquad 1.30$$

$$dt' = k\left(dt - \frac{v}{c^2}dx\right) \qquad 1.31$$

Then,

$$u'_x = \frac{dx'}{dt'} = \frac{dx - vdt}{dt - \frac{v}{c^2}dx}$$

$$\therefore u'_x = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} \qquad 1.32$$

$$u'_y = \frac{dy'}{dt'} = \frac{u_y\sqrt{1 - v^2/c^2}}{1 - \frac{u_x v}{c^2}} \qquad 1.33$$

$$u'_z = \frac{dz'}{dt'} = \frac{u_z\sqrt{1 - v^2/c^2}}{1 - \frac{u_x v}{c^2}} \qquad 1.34$$

Example 5

Two particles approach each other along parallel straight lines with equal velocities, each of magnitude 2 x 10^8 m/s. Find the relative velocity of one with respect to the other.

1.11 Mass Increase with Velocities

Consider ahypothetical collisionbetween two similar objects A and B. The masses of these objects relative to S are m_A and m_B. Their masses relative to S' (moving with uniform velocity with respect to S) arem_A' and m_B'. The object A is at rest in S and B is at rest in S'.

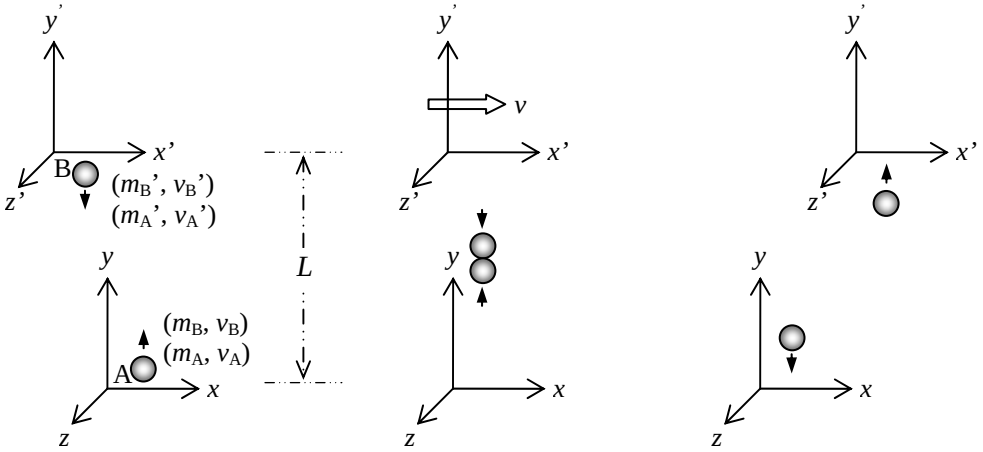


fig 1.4

The masses of A and B relative to S and S'respectively are the same; i.e., $m_A = m_B$' (= m_o, say.) Consider an elastic collision which occurs when they move along the *y*-direction with

velocities v_A and v_B relative to S, and v_A' and v_B' relative to S' such that $v_A = v_B$'. Such a collation is possible only when S' just passes by S so that their *y*-axes are in the same straight line. The collusion is supposed to occur when the objects have traveled equal distances along their *y*-axes.

Relative to S momentum conservation takes place:

$$m_A v_A + m_B(-v_B) = m_A(-v_A) + m_B v_B$$
$$\Rightarrow \quad m_A/m_B = v_A/v_B$$

If the distance between the *x*-axes of S and S' is *L*, the time for a round trip by A relative to S is:

$$t_o = \frac{L}{v_A}$$

Because $v_A = v_B$', the round trip by B relative to S' is the same as t_o:

$$t_o = \frac{L}{v'_B} = \frac{L}{v_A}$$

To an observer in S the time required by B would appear different, say *t*,

$$t = \frac{L}{v_B},$$

but $t = \dfrac{t_o}{\sqrt{1 - v^2/c^2}}$

therefore $\dfrac{t}{t_o} = \dfrac{1}{\sqrt{1 - v^2/c^2}} = \dfrac{v_A}{v_B} = \dfrac{m_B}{m_A} = \dfrac{m_B}{m'_B}$

hence, $m_B = \dfrac{m'_B}{\sqrt{1 - v^2/c^2}}$

In general, this can be written in the form:

$$m = \frac{m_o}{\sqrt{1 - v^2/c^2}} \qquad 1.35$$

The mass of a body moving at the speed *v* relative to an observer is larger than its mass when at rest relative to the observer by the factor $1/\sqrt{1 - v^2/c^2}$. This mass increase is reciprocal to an observer in S'.

The first verification of the increase in mass with velocity came from the works of Kaufmann in 1906 and of Bucherer in 1908. They discovered that the ratio *e*/*m* of the electron's charge to its mass is smaller for fast electrons than for slower ones.

When mass is treated as a variable, Newton's second law of motion becomes:

$$F = \frac{d}{dt}(mv) = m_o \frac{d}{dt}\left[\frac{v}{\sqrt{1-v^2/c^2}}\right]$$

1.12 Relativistic Mass-Energy Equivalence

Suppose a force F acting on a body of rest mass m_o through a distance dx for time dt produces an increase in its kinetic energy E_k by an amount dE_k. It acquires a uniform velocity v (= dx/dt) and a variable mass m. The change in kinetic energy is equal to the amount of work done. Hence,

$$\begin{aligned} dE_k &= Fdx = \frac{d}{dt}(mv)dx \\ &= m\left(\frac{dv}{dt}\right)dx + v\left(\frac{dm}{dt}\right)dx \\ &= m\left(\frac{dx}{dt}\right)dv + v\left(\frac{dx}{dt}\right)dm \\ &= mvdv + v^2 dm \end{aligned}$$

From the mass increase formula Eq. 1.35, we have $m^2\left(c^2 - v^2\right) = {m_o}^2 c^2$. Differentiating, we get:

$$mvdv + v^2 dm = c^2 dm$$

$$\therefore dE_k = c^2 dm \qquad 1.36$$

This shows that a change in E_k can be expressed in terms of mass as a variable. Whereas classically E_k could be increased only by increasing the velocity, here we have to take mass variation also into account. Integration of Eq. 1.36 gives,

$$\int_0^{E_k} dE_k = c^2 \int_{m_o}^{m} dm$$

$$\Rightarrow E_k = mc^2 - m_o c^2 \qquad 1.37$$

The following conclusions can be drawn frm Eq. 1.37:

i. The kinetic energy of a particle traveling with a velocity v is given by the equation:

$$\begin{aligned} E_k &= (m - m_o)c^2 = m_o c^2\left[\left(1 - v^2/c^2\right)^{-\frac{1}{2}} - 1\right] \\ &= \tfrac{1}{2} m_o v^2 + \tfrac{3}{8} m_o \frac{v^4}{c^2} + \ldots \quad \text{(power series)} \end{aligned}$$

The classical expression for $E_k (= \frac{1}{2} m_o v^2)$ gives only an approximate value for $v << c$.

ii. In Eq. 1.37 the term $(m_o c^2)$ is called the rest-mass-energy of the body E_o,

$$E_o = m_o c^2 \qquad 1.38$$

Clearly, this is the energy possessed by the mass of the body when it is (relatively) at rest. This means in addition to its kinetic, potential, electromagnetic, thermal, and other familiar guises, then, energy can manifest itself as mass.

The total energy (E) of the body can thus be written as the sum of its kinetic energy (E_k) and the rest-mass energy E_o.

$$E = E_k + E_o = mc^2 \qquad 1.39$$

The relation $E = mc^2$ forms the basis for understanding nuclear reactions such as fission and fusion. These reactions take place in nuclear bombs and nuclear reactors. The production of energy in stars and some other processes are understood today only as a result of the discovery of this important relation.

iii. Since mass and energy are not independent entities, the separate conservation principles of energy and mass are properly a single one, the *principle of conservation of mass-energy*. Mass can be created or destroyed but only if an equivalent amount of energy simultaneously vanishes or comes into being, and vice-versa.

Some Useful Formulas

From Eq. 1.35 and 1.39,

$$E^2\left(1-v^2/c^2\right) = {m_o}^2c^4$$

$$\Rightarrow E^2 = \left(\frac{E^2}{c^2}\right)v^2 + {m_o}^2c^4 = \left[\left(\frac{E}{c^2}\right)^2 v^2\right]c^2 + {m_o}^2c^4$$

Since momentum $p = mv = \dfrac{mc^2}{c^2}v = \dfrac{E}{c^2}v$

$$\Rightarrow E^2 = p^2c^2 + {m_o}^2c^4 \qquad 1.40$$

This is an extremely useful formula in relating the total energy of a particle to its momentum.

Other important relations in nuclear and elementary-particle physics are (*Exercise: show the relations.*)

$$p = m_o c\sqrt{\frac{1}{1-v^2/c^2}-1}$$

$$E_k = m_o c^2\left[\frac{1}{\sqrt{1-v^2/c^2}}-1\right]$$

$$\frac{v}{c} = \sqrt{1-\frac{1}{\left(1+E_k/m_oc^2\right)^2}}$$

$$\frac{1}{\sqrt{1-v^2/c^2}} = \sqrt{1+\frac{p^2}{m_o{}^2c^2}} = 1+\frac{E_k}{m_oc^2}$$

Particles of Zero Rest Mass

From Eq.1.40, the total energy for particles (e.g., photons) having no rest mass is $E = pc$. The energy of a photon is completely due to its motion and not at all to its rest energy. On being stopped they cease to exist- they are either absorbed completely or are changed into heat.

$$E = pc = mc^2, \text{ but } p = mv \quad \Rightarrow \quad mvc = mc^2$$

$$\Rightarrow v = c$$

This implies that a particle of zero rest mass must always move with the velocity of light in free space. Photons are not the only particles with zero rest mass. It is generally held now-a-days that gravitation is propagated with the velocity of light and quanta of gravitation, called gravitons, are particles with zero rest mass.

Unit of Measurement in Atomic and Nuclear Physics

- ***eV for Energy***

For a proton, with charge e = 1.602 x 10^{-19}C, being accelerated across a potential difference of 1 V, the work done is

$$W = qV = (1.602\times10^{-19}\,C)(1V) = 1.602\times10^{-19}\,J \equiv 1\,eV$$

$$\Rightarrow 1\,eV = 1.602\times10^{-19}\,J$$

Example 6

Calculate the rest energy of a proton.

- ***MeV/c^2 for Energy***

Since $E_o = m_oc^2 \Rightarrow m_0 = E_o/c^2$, the mass is often quoted in units of MeV/c^2; for example, the mass of the proton is given by 938 MeV/c^2.

Note: when we speak of a particle having certain energy, the common usage is to refer to the kinetic energy. A 6 GeV proton has a *kinetic energy* of 6 GeV, not a *total enery* of 6 GeV.

Example 7

Calculate the *v*, *p*, and *E* of a 2 GeV proton.

Example 8

Through what voltage should an electronhave to be acceleratedfrom rest so as to increase its mass by 0.4%? Find its speed under these conditions.

Example 9

A particle of rest mass m_o moving with a velocity of 0.6*c*collides inelastically with a similar particle at rest. Find the speed and rest mass of the composite particle if they coalesce after collision.

Exercises

1. How fast must a meter stick be moving if its length is observedto shrink to 0.500 m?
2. At what speed does a clock have to move if it is to beseen torun at a rate that is one-half the rate of a clockat rest?
3. The proper length of one spaceship is three times thatof another. The two spaceships are traveling in the samedirection and, while both are passing overhead, anEarth observer measures the two spaceships to have thesame length. If the slower spaceship is moving with aspeed of 0.350c, determine the speed of the fasterspaceship.
4. The pion has an average lifetime of 26.0ns when atrest. In order for it to travel 10.0 m, how fast must itmove?
5. A Klingon space ship moves away from the Earth at aspeed of 0.800c. The starship*Enterprise*pursues at a speed of 0.900c relative to the Earth. Observerson Earth see the *Enterprise* overtaking the Klingonship at a relative speed of 0.100c. With what speedis the *Enterprise* overtaking the Klingon ship as seen bythe crew of the *Enterprise*?
6. A spaceship moves away from the Earth at 0.500c andfires a shuttle craft in the forward direction at 0.500crelative to the ship. The pilot of the shuttle craftlaunches a probe at forward speed 0.500c relative to theshuttle craft. Determine (a) the speed of the shuttlecraft relative to the Earth and (b) the speed of theprobe relative to the Earth.
7. Calculate the momentum of an electron moving with aspeed of (a) 0.010 0c, (b) 0.500c, (c) 0.900c.
8. An unstable particle at rest breaks into two fragments of*unequal* mass. The mass of the lighter fragment is2.50×10^{-28} kg, and that of the heavier fragment is1.67×10^{-27}kg. If the lighter fragment has a speed of0.893c after the breakup, what is the speed of the heavierfragment?
9. The power output of the Sun is 3.77×10^{26} W. Howmuch mass is converted to energy in the Sun each second?
10. A gamma ray (a high-energy photon of light) canproduce an electron (e-) and a positron (e+) whenit enters the electric field of a heavy nucleus:$\gamma \rightarrow e^- + e^+$. What minimum γ-ray energy isrequired to accomplish this task? (*Hint*: The masses ofthe electron and the positron are equal.)